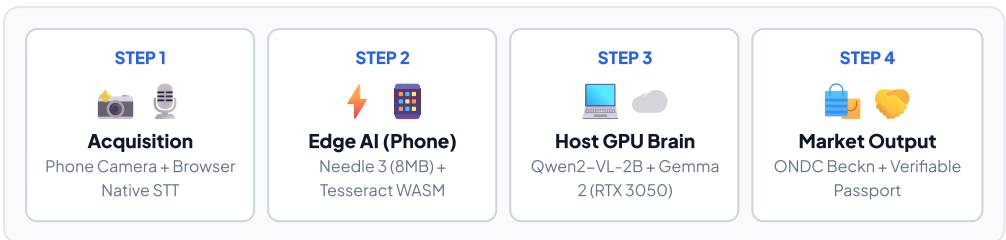


Technical Approach & Architecture

SIH 2026 • Ministry of Textiles • KalaSetu AI Platform Blueprint

SMART INDIA HACKATHON
2026

PROCESS FLOW DIAGRAM



WORKING PROTOTYPES

KalaSetu Smart Pricing ML

Cost Breakdown	
Material:	₹320
Labor (8h):	₹480
Overhead:	₹40

SUGGESTED PRICE
₹840
Vs. middleman: ₹350
(-58%)

KalaSetu Voice Catalog LIVE

Blue Pottery Kalash

"Yeh neeli mitti ka kalash hai..."

MICRO-TEXTURE VLM OPTIMIZATION (RTX 3050 4GB VRAM)			GPU ALLOCATION: ~1.4 - 2.2 GB
Model	Scale	Resolution Strategy	DocVQA / TextVQA
Qwen2-VL-2B-Instruct	2.21B	Dynamic Native Resolution	90.1% / 79.7% 🏆
PaliGemma-2-3B	2.92B	Fixed (448 x 448)	84.5% / 74.1%
SmoVLM-Instruct	2.20B	Dynamic Patch Tiling	81.6% / 72.7%
Moondream2	1.86B	Fixed (378 x 378)	70.5% / 65.2%

UPDATED 2026 TECH STACK

Frontend: React.js + Tailwind CSS / Lightweight PWA (< 150 MB RAM on budget phones) packaged into Android APK via PWABuilder (TWA).

Backend & Cloud BaaS: Supabase (Free Tier) — Auth, Postgres DB, Media Storage, and pgvector for vector visual search.

Edge AI & On-Device Automation: Needle 3 (8-29 MB) (Cactus Compute) for local tool calling & field extraction + Jev (TypeSafe AI) sub-100ms claim risk validation + TFLite (U2Net) + Tesseract.js WASM for offline slip OCR.

Micro-Texture Vision (VLM): Qwen2-VL-2B-Instruct (90.1% DocVQA SOTA) + PaliGemma-2-3B on NVIDIA RTX 3050 CUDA (4GB VRAM). Identifies weave defects, motifs, and Pehchan IDs.

Reasoning & Story Brain: Google Gemma 2 (2B / 9B) / Qwen2.5-3B for structured catalog generation, emotional backstories, and anti-exploitation bargaining logic.

Voice AI & Multilingual STT: Browser Native STT (Web Speech API, 0 MB overhead) + Bhashini API (Govt of India, 22 languages) + Sarvam AI / ElevenLabs Indic voice.

Pricing Engine: Scikit-learn ML Regression + FairPrice Guard cost-plus wage defense.

Logistics & Provenance Maps: Leaflet.js + OpenStreetMap with 378 SFURTI cluster geo-fence polygons.

Marketplace Integration: ONDC Beckn Protocol + DLT SMS Gateway (Fast2SMS / MSG91).

DATASETS GROUNDING THE MODEL:

- 4th All India Handloom Census: 35.22 Lakh workers wage & cluster data.
- EPCH Annual Report 2024-25: ₹33,424 Cr handicraft export valuation.
- SFURTI Cluster Directory (MSME): 378 recognized rural artisan clusters.
- Kaggle Handmade Products & Indo Fashion: Pricing model training set.
- Bhashini AI Mission: 22 Indian regional languages speech corpus.